

Write xml dtd xsd and xslt to render in browser and also write xquery.

XML:

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl"
href="books.xsl"?>
```

```
<!DOCTYPE bookstore SYSTEM "books.dtd">
<bookstore
xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
```

```
xsi:noNamespaceSchemaLocation="books.xsd"
>
```

```
<book category="cooking">
  <title lang="en">Everyday Italian</title>
  <author>Giada De Laurentiis</author>
  <year>2005</year>
  <price>30.00</price>
</book>
```

```
<book category="web">
  <title lang="en">Learning XML</title>
  <author>Erik T. Ray</author>
  <year>2003</year>
  <price>39.95</price>
</book>
```

```
<book category="fantasy">
```

Books.xsd

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
xmlns:xs="http://www.w3.org/2001/XMLSchema-
ma">
```

```
<xs:element name="bookstore">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="book"
maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="title">
              <xs:complexType>
                <xs:simpleContent>
                  <xs:restriction base="xs:string">
                    <xs:attribute name="lang"
type="xs:language" use="required"/>
                  </xs:restriction>
                </xs:simpleContent>
              </xs:complexType>
            </xs:element>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
```

```
<title lang="en">Harry Potter</title>
<author>J K. Rowling</author>
<year>2005</year>
<price>69.99</price>
</book>
```

```
<book category="children">
  <title lang="en">The Little Prince</title>
  <author>Antoine de Saint-Exupéry</author>
  <year>1943</year>
  <price>85.50</price>
</book>
```

```
</bookstore>
```

Books.dtd:

```
<!ELEMENT bookstore (book*)>
<!ELEMENT book (title, author, year, price)>
<!ATTLIST book category CDATA
#REQUIRED>
```

```
<!ELEMENT title (#PCDATA)>
<!ATTLIST title lang CDATA #REQUIRED>
```

```
<!ELEMENT author (#PCDATA)>
<!ELEMENT year (#PCDATA)>
<!ELEMENT price (#PCDATA)>
```

```
</xs:element>
<xs:element name="author"
type="xs:string"/>
<xs:element name="year"
type="xs:gYear"/>
<xs:element name="price"
type="xs:decimal"/>
</xs:sequence>
<xs:attribute name="category"
type="xs:string" use="required"/>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:schema>
```

```

Books.xsl
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Trans
form">
  <xsl:output method="html" indent="yes"
encoding="UTF-8"/>
  <xsl:template match="/">
    <html> <head>
      <title>Bookstore - Books with price &gt;
50.00</title>
      <style>
        body { font-family: Arial, sans-serif;
margin: 40px; }
        h1 { color: #2c3e50; }
        table { width: 80%; border-collapse:
collapse; margin-top: 20px; }
        th, td { border: 1px solid #bdc3c7;
padding: 10px; text-align: left; }
        th { background-color: #ecf0f1; }
        tr:nth-child(even) { background-color:
#f9f9f9; }
      </style> </head>
      <body>
        <h1>Books with price greater than
$50.00</h1>
        <table> <tr>
          <th>Title</th>
          <th>Author</th>
          <th>Year</th>

```

```

          <th>Price</th>
          <th>Category</th>
        </tr>
        <!-- This is the key part: select only books
where price > 50 -->
        <xsl:for-each
select="bookstore/book[price > 50]">
          <xsl:sort select="price"
order="descending" data-type="number"/>
          <tr>
            <td><xsl:value-of select="title"/></td>
            <td><xsl:value-of
select="author"/></td>
            <td><xsl:value-of select="year"/></td>
            <td><xsl:value-of
select="price"/></td>
            <td><xsl:value-of
select="@category"/></td>
          </tr>
        </xsl:for-each>
      </table>
      <p style="margin-top:30px;
color:#7f8c8d;">
        Total books shown: <xsl:value-of
select="count(bookstore/book[price > 50])"/>
      </p>
    </body>
  </html>
</xsl:template>
</xsl:stylesheet>

```

file:///home/prajwal/Desktop/Lab_reports/fourth_sem/WEBTECH-I/xml_bookstore/books.xml

Books with price greater than \$50.00

Title	Author	Year	Price	Category
The Little Prince	Antoine de Saint-Exupéry	1943	\$85.50	children
Harry Potter	J K. Rowling	2005	\$69.99	fantasy

← → ↻ xpather.com ☆ ⓘ ⌵

(:XQuery:- Prajwal :) Elements found: 2

/bookstore/book[price > 50]

XML mode Format Copy Text Node

```

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<!DOCTYPE bookstore SYSTEM "books.dtd">
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xsi:noNamespaceSchemaLocation="books.xsd">
  <book category="fantasy">
    <title lang="en">Harry Potter</title>
    <author>J K. Rowling</author>
    <year>2006</year>
    <price>69.99</price>
  </book>

  <book category="children">
    <title lang="en">The Little Prince</title>
    <author>Antoine de Saint-Exupéry</author>
    <year>1943</year>
    <price>85.50</price>
  </book>

  <book category="cooking">
    <title lang="en">Everyday Italian</title>
    <author>Giada De Laurentiis</author>
    <year>2005</year>
    <price>30.00</price>
  </book>
</bookstore>

```

1. Harry Potter J K. Rowling 2006 69.99
2. The Little Prince Antoine de Saint-Exupéry 1943 85.50

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